Ziyang Ye

yeziyang777@gmail.com | GitHub

Research Interests

Generative video modeling, interactive world models, multimodal learning, and foundation models for embodied intelligence, with the broader goal of connecting visual prediction, world dynamics, and action.

Education

- **Jilin University** — *Sep. 2023 - Jun. 2027*
  Bachelor of Engineering in Software Engineering — Changchun, China
  - **GPA:** 88.68/100

Publications

[C.1] Junchao Huang, **Ziyang Ye**, Xinting Hu, Tianyu He, Guiyu Zhang, Shaoshuai Shi, Jiang Bian, Li Jiang. **LIVE: Long-horizon Interactive Video World Modeling**. *International Conference on Machine Learning (**ICML**)*, 2026.

Research Experience

- **The Chinese University of Hong Kong, Shenzhen (CUHK-SZ)** — *Dec. 2025 - Present*
  Research Assistant (Advisor: Prof. Li Jiang) — Shenzhen, China
  - **LIVE** (ICML 2026): Contributed to implementing the cycle-consistency training loop for a Causal DiT, including forward-rollout scheduling, temporal reversal of frames and controls, reverse-causal masking, and flow-matching loss integration.
  - Conducted training and evaluation on RealEstate10K, UE Engine, and Minecraft using PSNR, SSIM, LPIPS, FID, and FVD, including analysis beyond the training rollout horizon.
  - **GeometryWAM**: Designed a world-action DiT for embodied manipulation using privileged depth and 3D targets during training, with asymmetric attention masking policy queries from geometry inputs at deployment.
  - **Multi-Player Interactive World Model** (ongoing): Exploring a multi-player extension using scene-time synchronization, cross-player communication, shared latent memory, and private-view decoding to study cross-view consistency under simultaneous actions.
- **School of Artificial Intelligence, Jilin University** — *Mar. 2025 - Oct. 2025*
  Research Intern (Advisor: Prof. Xi Yang) — Changchun, China
  - Built a sketch-conditioned appearance-transfer pipeline integrating FLUX/ControlNet generation, SAM-based spatial guidance, and DINO feature alignment.
  - Designed style-content disentanglement and evaluated transfer fidelity using DINO, CLIP-I, and SSIM.

Internship Experience

- **Didi, Voyager AI Research** — *Jun. 2026 - Present*
  Intern (Mentor: Shaoshuai Shi) — Guangzhou, China
  - Developing Vision-Language-Action (VLA) systems for complex real-robot manipulation tasks.

Technical Skills

- **Programming & Training:** Python, C/C++, PyTorch, Distributed Data Parallel (DDP), Hugging Face Diffusers
- **Generative & World Models:** Diffusion Transformers (DiT), Flow Matching, ControlNet, Wan, FLUX, Interactive Video World Models, World Action Models (Cosmos-Policy, Fast-WAM, Lingbot-VA)
- **Vision, Multimodal & Embodied:** Vision Transformers (ViT, DINO), Vision-Language Models (CLIP, Qwen), 3D Scene Understanding (VGGT/Pi3), Segmentation (SAM), Vision-Language-Action Models (Pi0, OpenVLA)
- **Systems & Tools:** Linux, Git, Docker, tmux, OpenCV, FFmpeg, Weights & Biases, SwanLab, LaTeX

Honors and Awards

- **Honorable Mention, Mathematical Contest in Modeling (MCM)** — *2026*
  COMAP
- **Academic Scholarship & Second-Class Scholarship** — *2025*
  Jilin University
- **Outstanding Student Leader & Second-Class Scholarship** — *2024*
  Jilin University